

外

平成18年度 筑波大学大学院 博士課程 システム情報工学研究科
コンピュータサイエンス専攻 入学試験問題
外国語

英語

問題 次頁の英文を読んで、問(1)(2)は1枚目、問(3)は2枚目に回答しなさい。

Problem Read the English writing on the next page, and then answer the questions (1)(2) and (3) in separate answer sheets, respectively.

One day in the 1960's, while shaving in front of a mirror, psychologist Gordon Gallup wondered what would happen if you put animals in front of a looking glass. He created what he called the mark test, in which primates already familiar 1 mirrors and their own reflection were anesthetized and marked with an odorless dye above an eyebrow and on an ear. When the animals woke up, they were shown their reflection in the mirror.

Two species, chimpanzees and orangutans, reacted by touching the dyed spots – evidence, Gallup argued, that these species are self-aware. ①But Donald Griffin, a biologist at Harvard University, thinks other species are also self-aware, even if they aren't passed the mark test. It doesn't make sense, says Griffin, that chimps and oranges succeed at the test, while other apes, like gorillas, do not. 2 , only 75 percent of adult chimps pass the test. "You'd have to say that some chimps are self-aware and others aren't, which seems a bit ridiculous," Griffin says.

Some researchers contend that dolphins can pass a variation of Gallup's mark test. Hardly anything in this field is 3 of controversy. The stakes are high: At issue is the degree to which humans are different from other 4 . Are we special, or just ... conceited?

②Daniel Povinelli, a cognitive scientist at the University of Louisiana, argues that even small children who pass the mark test lack the kind of self-awareness that older children have. In one of Povinelli's experiments, kids watch a video of someone surreptitiously placing a large, brightly colored sticker on top of their heads just minutes earlier. Most three-year-olds, upon seeing the video, fail to reach up and remove the sticker. They recognize themselves ("That's me!") but don't quite grasp that the sticker is still on their heads ("That sticker's on his head"). And yet most four-years-olds pass the test.

③It's hard to say for certain what the mark test really measures. Research on consciousness and self-awareness is dogged by ambiguity. Is it possible that some animals, for example, fail the test simply because they lack the motor skills to touch the mark? Sometimes in science we get interesting answers – but can't quite decide what it was that we asked.

(Joel Achenbach, *National Geographic*, January 2004, から抜粋)

注： primate 霊長類
anesthetize 麻酔をかける
dye 染料
orangutan オラウータン
chimp チンパンジー
ape 類人猿

stakes 賭け, 危険率
conceit うぬぼれ
cognitive scientist 認知科学者
surreptitiously そっと秘密に
dog つきまとう

問（１） 下線部①②③を和訳せよ。

Question (1) Translate the underlined parts ①, ② and ③ into Japanese.

問（２） ～ にあてはまる単語を以下から選びアルファベットで答えなさい。

Question (2) Choose the most suitable word for blanks ～ from the list given below, and answer with the character.

A. although

B. besides

C. creatures

D. free

E. of

F. out

G. reptiles

H. with

問（３） Daniel Povinelli が行なった実験とその結果を要約して，簡潔に日本語または英語で記述しなさい。

Question (3) Summarize the experiment, including its results, that was carried out by Daniel Povinelli, and describe shortly and clearly in either Japanese or English.